

MMA 616: Managing Intelligence

Course Overview

Department:	Strategy, Entrepreneurship, and Management
Professor:	Dr. Vern L. Glaser (vglaser@ualberta.ca)
Teaching Assistant:	Jennifer Kotadia (jkotadi1@ualberta.ca)
Dates:	June 26, 27 & July 3, 4, 2026
Time:	9:00 AM – 5:00 PM
Location:	ESB-236
Office Hours:	Upon request

MMA students arrive with strong technical foundations in machine learning, statistics, and AI applications. Technical fluency, however, does not by itself create organizational value. Most AI initiatives stall not because the technology fails but because organizations cannot identify the right opportunities, build solutions that integrate with existing workflows, or sustain adoption beyond an initial pilot. For a technically fluent room, the binding constraint is not capability but **taste and opportunity-selection**: judgment about what is worth building and what “good” output looks like.

This course is a four-day **group hackathon**. Working in teams, students develop that judgment and then build, deploy, and evaluate a **data-based application with an agentic intelligence layer** (a dashboard or agent on a data source of the group’s choosing), working entirely in Claude Code, with a CLAUDE.md file as the project’s living specification. Over four intensive days with a one-week deployment gap, each group defines an objective, builds a working prototype, deploys it in a real setting, and evaluates what happened when design met practice. A take-home final then asks each student to apply the full method independently, building their own application on a Government of Alberta open dataset.

Day 1: Developing the Goal & Objective

Friday, June 26

Day 1 establishes a method for deciding *what* is worth building. For a room that can already build, value is gated by judgment: a “taste” for what a genuinely useful data product looks like, and the discipline to recognize what a data source can and cannot responsibly support. Through a worked example, a dashboard application with an intelligence layer, students learn to read a dataset for opportunity, distinguish high-value uses from technically-correct-but-useless ones, and define success criteria concrete enough to evaluate later. Groups form, choose a data source (their own or synthetic), and converge on a single application concept. By day’s end, each group has authored the project’s CLAUDE.md, the living specification that drives the build.

Day 2: Developing the Agentic Infrastructure

Saturday, June 27

Day 2 turns the specification into a deployed prototype. Working in Claude Code, groups build the agentic infrastructure behind their application: packaged skills, connections to their data source, and dynamic workflows that orchestrate multi-step analysis. The day emphasizes strategic build discipline: designing for the person who will use the output, scoping to what can ship in a week, and judging when to let the agent decide versus hard-code a reliable pipeline. Groups also establish a lightweight evaluation baseline so that improvement can be measured later. By session’s end, each group has a working prototype deployed for real-world use.

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Between Days 2 and 3, groups deploy their application in real work for one week, maintaining a structured log of what worked, what surprised them, where they intervened, and how others reacted.

Day 3: Evaluating the Agentic Infrastructure

Friday, July 3

Day 3 begins with deployment stories: each group shares what happened when its application met real work. The governing insight is that building an application is not the same as integrating one. Recalling the Three Levels framework from MMA 608, now run reflexively on the group's own application, students examine the gap between the behavior they specified and the behavior the application actually enacted, a gap that reveals priorities they did not know they held. The day provides a structured diagnostic for evaluating output quality across workflow, decision rights, metrics, and override patterns, and groups build and run an evaluation harness that produces before-and-after measurement of their improvements. Each group produces a deployment assessment and improvement plan that anchors the Day 4 capstone.

Day 4: Capstone Presentations & Individual Launch

Saturday, July 4

Day 4 is the course climax. Each group presents its capstone, a live demonstration of its deployed application, with the strategic rationale, technical execution, and before-and-after evaluation evidence that justify it, and defends it under questioning. A closing synthesis frames the enduring output of the course: not the application itself, but a repeatable method for creating value with AI, built by developing an objective, building agentic infrastructure, and evaluating what it reveals. The day ends by launching the individual take-home final, in which each student applies that method independently.

Assessment Components

Component	Weight	Description
Class Participation	20%	Active engagement in discussions, hackathon work, deployment stories, and peer feedback across all four days.
Group AI Project	40%	A team-built data-based application with an agentic intelligence layer, specified, built, deployed, evaluated, and presented by the group across all four days.
Final Exam	40%	Take-home individual exam: build your own data-based application on a Government of Alberta open dataset, applying the course's full method from objective and specification through agentic infrastructure to evaluation. Distinct from the group project.

Alberta School of Business Competency Goals. This course addresses the Alberta School of Business Competency Goals (updated August 2024): (1) *Business Concepts and Theories*, through the agentic-AI value-creation method, the Axiology Three-Levels framework, and the strategic-fit and evaluation frameworks; (2) *Critical Thinking*, through opportunity-selection under ambiguity, the structured agent audit, and the design of evaluation harnesses that test what “good” output means; (3) *Professional Skills*, through the team-based build and deployment and the capstone presentation and defense; and (4) *Responsible Business Principles*, through responsible data use, the evaluation of output quality and bias, and the examination of the gap between the values an application is configured with and the values it enacts in practice.

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Required Readings

The readings for this course combine curated reference guides with foundational texts and contemporary sources.

Course Reference Guides. Comprehensive teaching guides developed for each major course topic, synthesizing academic research, industry frameworks, and practical applications. Distributed via Canvas before each session.

Foundational Text. The following book provides the theoretical foundation for our organizational and strategic discussions:

- Mollick, E. (2024). *Co-Intelligence: Living and Working with AI*. Penguin Random House.

Contemporary Sources. AI capabilities and best practices evolve weekly. Links to current blogs, Substacks, research papers, and industry reports are shared throughout the course via eClass.

AI in This Course

AI tools are integral to this course: you will use them to build your application, research your strategy, and accelerate your project work. Students are encouraged to use AI throughout. But strategic judgment about where AI creates value, how to evaluate what your application reveals about your priorities, and how to lead its integration requires the kind of professional discernment that comes from deployment experience, not from AI-generated analysis. Use AI to extend your reach. Maintain your own judgment about what it produces.

Formal Policy Statements

Policy about course outlines can be found in [Course Requirements, Evaluation Procedures and Grading](#) of the University Calendar.

Grading System. Grade distribution will follow University of Alberta guidelines. As a graduate course, MMA 616 uses the graduate grading scale, described in the University Calendar under [Evaluation Procedures and Grading System](#).

Missed Assignments. Approval for an excused absence from term work is at the instructor's discretion. Any student incapacitated because of illness, suffering from severe domestic affliction, or having other compelling reasons (including religious conviction) may apply for an excused absence for a missed assignment. If you have a conflict, please discuss it with me **beforehand**, and I will be happy to work out a solution with you. In fairness to other students, I will rarely accept excuses once an assignment deadline has passed.

Appropriate Conduct. My goal in this course is to create a supportive environment for learning based on open, constructive debate. This requires all of us to be engaged with the material and with each other in a professional manner, with courtesy and respect for each other's individuality. Discrimination, malicious criticism, and disruption of class are examples of conduct that are not acceptable.

Recording Policy. Audio or video recording, digital or otherwise, of lectures, labs, seminars or any other teaching environment by students is allowed only with the prior written consent of the instructor or as a part of an approved accommodation plan. Student or instructor content, digital or otherwise, created and/or used within the context of the course is to be used solely for personal study, and is not to be used or distributed for any other purpose without prior written consent from the content author(s).

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Academic Integrity. The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the [Student Academic Integrity Policy](#) and avoid any behaviour that could potentially result in suspicions of cheating, plagiarism, misrepresentation of facts, and/or participation in an offence. Academic dishonesty is a serious offence and can result in suspension or expulsion from the University. I will immediately forward any suspicious cases to the appropriate Associate Dean. *Plagiarism will lead to evaluating assignments as “fail” (0%). Plagiarism can also lead to a student failing the course.*

Other Resources. If you have any personal concerns, the following support numbers may be helpful resources:

- Support Network, Crisis Support: 780-482-4357
- Community Urgent Services and Stabilization: 780-342-7777